



Chandrayaan-2

OHRC PDS4 Data Products

User Guide



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Chandrayaan-2 OHRC PDS4 Data Products User Guide

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1. Introduction

1.1 Purpose and Scope

The purpose of this user guide document is to bring about a details about the Chandrayaan-2 PDS4 Data Products from the user's perspective. It lists out the number of data products, types and formats available for the users to work with PDS4 data products.

1.2 Audience

This report is meant for the staff of archiving authority (ISSDC for ISRO), data processing team, PI team, and users from scientific community.

1.3 Definitions, Acronyms & Abbreviations

Table 1 brings out the list of definitions, acronyms & abbreviations used in this report.

Table 1 Definitions, Acronym & Abbreviation

Acronym	Abbreviations/Description
PDS	Planetary Data System
ISRO	Indian Space Research Organization
ISSDC	Indian Space Data Centre
ISDA	ISRO Science Data archive
SAC	Space Applications Centre
DP	Data Processing
PI	Principal Investigator

1.4 Applicable Documents

1. Chandrayaan-2 Science data management and Archive plan (Chandryaan-2/DP/SAC/SIPG/HRDPD/TR-06/July 2018)
2. Chandrayaan-2 SAC DP – ISSDC Interface for Archival (Chandrayaan-2/DP/SAC/SIPG/HRDPD/TR-13/Dec. 2018)

2. Chandrayaan-2 PDS4 Data Archive

Chandrayaan-2, India's second mission to the Moon was launched by GSLV-MK-III in July 2019. Mission carries 13 scientific payloads on Orbiter. Orbiter is presently orbiting moon under 100km x 100 km circular orbit.

The data collected from the instruments from the orbiter will be received, processed and archived in Indian Space Science Data Center (ISSDC) for dissemination and use by scientific community in India and abroad.

The data from Chandrayaan-2 instruments should be archived in such a way that they are easily accessible by Internet through simple interfaces being used elsewhere for similar kind of missions.

PDS4 is the de facto international standard for long term archival of planetary science data. For all the scientific payloads, PDS4 data products types [1] were identified and shown below in the **table 2**

Table 2 Chandrayaan-2 PDS4 Data Products Types

Sr. No.	Instruments	PDS4 Data Products Types		
		Raw	Calibrated	Derived
1	TMC2	Yes	Yes	Yes
2	IIRS	Yes	Yes	Yes
3	OHRC	Yes	Yes	No
4	SAR	Yes	Yes	No
5	CLASS	Yes	No	No
6	XSM	Yes	Yes	No
7	CHACE-2	Yes	No	No

Each of the PDS4 data products types were mapped to the ISRO Levels and CODMAC Level as shown in **table 3**.

Table 3 Chandrayaan-2 PDS4 CODMAC Mapping

PDS4 Level	PDS4 Level Description	ISRO Level	CODMAC Level
Raw	Original data from an instrument after initial pre-processing like decompression, reformatting, packetization etc. the archived data are in a PDS approved archive format.	0	2
Calibrated	Data converted to physical units, which makes values independent of the instrument (Radiometrically corrected & seleno-tagged).	1	3
Derived	Results that have been distilled from one or more calibrated data products (for example, maps, gravity or magnetic fields, or ring particle size distributions). Supplementary data, such as calibration tables or tables of viewing geometry, used to interpret observational data should also be classified as 'derived' data if not easily matched to one of the other three categories (higher level products like ortho, DEM).	2	4+

3. Archive Structure

Archive structure is designed both at instrument level and data products level.

3.1 Instruments Collection and Data Products

Next level instrument wise collection is defined based on payload name for example for OHRC payload, ohr_collection is defined. Under each instrument wise data collection, data is organized under each directory defined by PDS4. Data is organized based on the levels of data processing and data products definitions. All the data is planned to store as Year and month wise. **Figure 1** shows detail instruments wise collection structure [2].

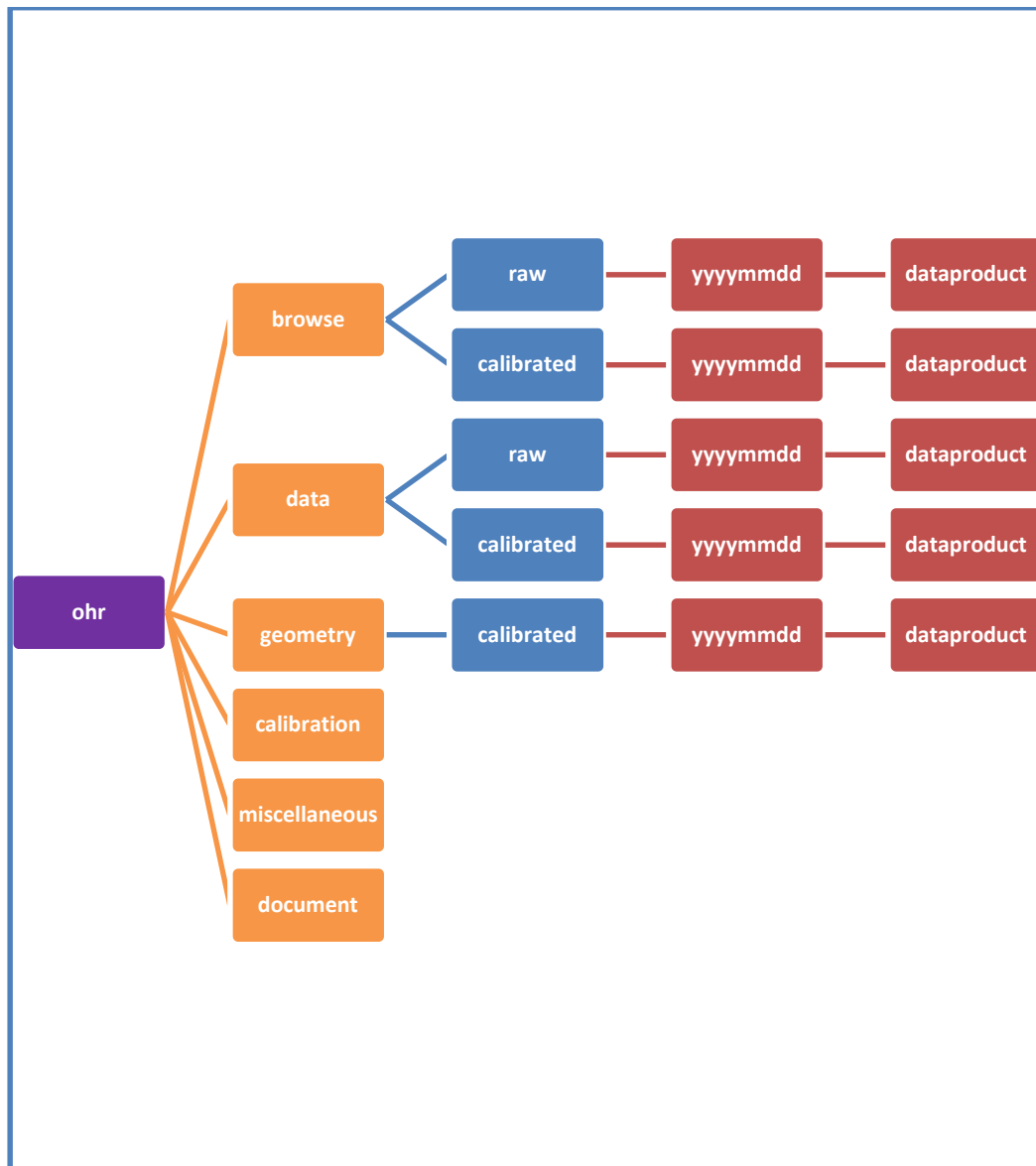


Figure 1 Instrument Collection

4. OHRC PDS4 Data Products

The PDS4 Data Products for OHRC defined based on the defined data processing levels (L0 and L1) as Raw and Calibrated data products. more details shown in next sub sections.

3.1 Raw Data Products

Raw Data Products contains raw data without any correction along with the system level corner coordinates. For each payload observation, PDS4 data products are generated (based on product id derived from file name (Annexure I)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in figure2.

For example, user has downloaded the zip product as ch2_ohr_nrp_20190906T224128571429200_d_img_gds.zip.

Figure 2 shows unzipped structure of OHRC Raw Data Product zip file at user end.

```
data
├── raw
│   └── 20190906
│       ├── ch2_ohr_nrp_20190906T224128571429200_d_img_gds.img
│       └── ch2_ohr_nrp_20190906T224128571429200_d_img_gds.xml
browse
├── raw
│   └── 20190906
│       ├── ch2_ohr_nrp_20190906T224128571429200_b_brw_gds.png
│       └── ch2_ohr_nrp_20190906T224128571429200_b_brw_gds.xml
miscellaneous
├── raw
│   └── 20190906
│       ├── ch2_ohr_nrp_20190906T224128571429200_d_img_gds.lbr
│       ├── ch2_ohr_nrp_20190906T224128571429200_d_img_gds.oat
│       ├── ch2_ohr_nrp_20190906T224128571429200_d_img_gds.oath
│       └── ch2_ohr_nrp_20190906T224128571429200_d_img_gds.spm
```

Figure 2 OHRC Raw Data Archive

Table 4 shows the details of individual data products (with the data formats) present inside the zip product. To visualize the data products please refer Annexure II

Table 4 OHRC Raw Data Products

PDS4 Data Product Levels	PDS4 Data Products	Data Formats	Description	Visualization tools available
For sensor OHRC - Pan Raw	Data Raw Image	Binary	The PDS Raw data in generic binary	Any binary image viewer – imageJ, envi, erdas Provide image width, image height and data type from label file
	Data Raw Label	XML	The PDS Label Product contains description about the Raw Product.	PDS4 Viewer
	Browse Raw Image	PNG	Sub sampled browse raw image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse Raw Label	XML	The PDS Label Product contains description about the Browse Raw Product.	
	Miscellaneous Data (non PDS data)	Text	Contains four files Orbit & Attitude (oat) Orbit & Attitude header(oath) Liberation angle (lbr) Sun parameter file spm	Any text viewer software will be able to view the data

3.2 Calibrated Data Products

Calibrated Data Products contains radiometrically corrected data along with the refined corner coordinates. For each payload observation, PDS4 data products are generated (based on product id derived from file name (Annexure I)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in figure2. For example, user has downloaded the zip product as ch2_ohr_ncp_20190906T224128571429200_d_img_gds.zip. **Figure 3** shows unzipped structure of OHRC Calibrated Data Product zip file at user end. To visualize the data products please refer Annexure II.

```
data
├── calibrated
│   └── 20190906
│       ├── ch2_ohr_ncp_20190906T224128571429200_d_img_gds.img
│       └── ch2_ohr_ncp_20190906T224128571429200_d_img_gds.xml
browse
├── calibrated
│   └── 20190906
│       ├── ch2_ohr_ncp_20190906T224128571429200_b_brw_gds.png
│       └── ch2_ohr_ncp_20190906T224128571429200_b_brw_gds.xml
geometry
├── calibrated
│   └── 20190906
│       ├── ch2_ohr_ncp_20190906T224128571429200_g_grd_gds.csv
│       └── ch2_ohr_ncp_20190906T224128571429200_g_grd_gds.xml
miscellaneous
├── raw
│   └── 20190906
│       ├── ch2_ohr_ncp_20190906T224128571429200_d_img_gds.lbr
│       ├── ch2_ohr_ncp_20190906T224128571429200_d_img_gds.oat
│       ├── ch2_ohr_ncp_20190906T224128571429200_d_img_gds.oath
│       └── ch2_ohr_ncp_20190906T224128571429200_d_img_gds.spm
```

Figure 3 OHRC Calibrated Data Archive

Below table shows the details of individual data products (with the data formats) present inside the zip product.

Table 5 OHRC Calibrated Data Products

PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Description	Visualization tools available
For sensor OHRC - Pan Calibrated	Data Calibrated Image	Binary	The PDS Radio metrically corrected data in generic binary format	Any binary image viewer – imageJ, envi, erdas Provide image width, image height and data type from label file
	Data Calibrated Label	XML	The PDS Label Product contains description about the Calibrated Product.	PDS4 Viewer
	Browse Calibrated Image	PNG	Sub sampled browse calibrated image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse Calibrated Label	XML	The PDS Label Product contains description about the Browse Calibrated Product.	
	Geometric Calibrated Grid	CSV	A grid file containing lon, lat, scan and pix as four columns along with the data	Any text editor, ms excel can be used to load the csv file.
	Geometric Calibrated Grid Label	XML	The PDS Label Product contains description about the Geometry Calibrated Product.	PDS4 Viewer
	Miscellaneous Data (non PDS data)	Text	Contains four files Orbit & Attitude (oat) Orbit & Attitude header(oath) Liberation angle (lbr) Sun parameter file spm	Any text viewer software will be able to view the data

5. Archive Products formats

Data that comprises the OHRC archives are formatted in accordance with PDS specifications. This section provides details on the formats used for each of the products included in the archive. The detailed file formats for the defined data products based on PDS4 data types have been defined shown in next sub section.

5.1 Data File Formats

OHRC instrument is an imaging payload. The primary data collected from instrument is raw image. The OHRC imager has panchromatic (PAN) sensor. For each sensor both primary as well as secondary data sets are defined that contains data product levels, types and formats shown in the table 6.

Table 6 OHRC PDS4 Data Products

Sensors	PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Data type*
For high resolution panchromatic sensor	Raw	Image	Binary	UnsignedByte
	Raw	Browse	PNG	UnsignedByte
	Calibrated	Image	Binary	UnsignedByte
	Calibrated	Browse	PNG	UnsignedByte
	Calibrated	Geometry	CSV	ASCII Text

Note*:

UnsignedByte is Byte (Unsigned byte of 8-bit length)

6. Local Data Dictionary

Every mission has some conditions based on the events that needs to be handle as an information at output product level. Some information is common to every planetary mission launched by ISRO. There is an ISRO Science Data Archive established during Chandrayaan-1 mission in 2008. All the products hosted for users under ISDA are PDS compliance. For Chandrayaan-2 mission, PDS4 Archive standard was adopted. PDS4 is currently next and future generation archive standard adopted by all the space agencies including NASA, ESA and JAXA for their current and future mission. To handle the ISRO planetary mission's specific parameters, there is a concept in PDS4 called Local Data Dictionary i.e. LDD. In LDD one can define all the parameters and that can be used as a reference for future planetary missions in India. LDD takes basic data dictionary of PDS4 and then top of it builds local dictionary. For Chandrayaan-2 mission and future planetary missions of ISRO, list of parameters is identified shown in details in below table.

Table 7 ISDA Local Data Dictionary Class and Attributes List

Class/Sub Class Hierarchy	Attribute Name	Description/Value
Mission_Area Product_Parameters	job_id	Unique id assign by DP Scheduler when the product is ingested
	level0_dir_name	DP Input Level-0 directory name
	imaging_orbit_number	Imaging Orbit Number assigned on board
	dumping_orbit_number	Dumping Orbit Number is assigned based on the data collected on ground station
	line_exposure_duration	Image integration time
	bits_selection	Bit selection based on TDI settings. It can have values like lsb, msb, mid
	tdi_stages	Time delay integration stages
	detector_pixel_width	Pixel width of detector
	focal_length	Instrument focal length
	spacecraft_altitude	Spacecraft height
	orbit_limb_direction	Orbit limb direction it can have two values - Ascending or Descending Spacecraft yaw mode it can
	spacecraft_yaw_direction	

		have two values – True/False a) True means Reverse, in flight yaw mode Reverse describes 180 degree rotated from nominal. b) False means Forward, in flight yaw mode Forward describes 0 degree rotated from nominal) There are four possible orbit_limb_direction and spacecraft_yaw_direction combinations: 1. Descending and False, no changes in lines or samples are performed 2. Descending and True, samples are reversed 3. Ascending and False, both lines and samples are reversed 4. Ascending and True. Lines are reversed
	reference_data_used	Reference data set used in refining the system level corners
	pixel_resolution	Image pixel resolution expressed in unit m/pixel
Mission_Area • Geometry_Parameters ○ System_Level_Coordinates ○ Refined_Level_Coordinates	upper_left_latitude	Upper left latitude reticle point with number 1
	upper_left_longitude	Upper left longitude reticle point with number 1
	upper_right_latitude	Upper right latitude reticle point with number 3
	upper_right_longitude	Upper right longitude reticle point with number 3
	lower_left_latitude	Lower left latitude reticle point with number 7
	lower_left_longitude	Lower left longitude reticle point with number 7
	lower_right_latitude	Lower right latitude reticle point with number 9
	lower_right_longitude	Lower right longitude reticle point with number 9

7. Conclusion

This user guide is abstract view of PDS4 Data Products. As per the mission planning, and operations plan, data was collected from the onboard payload at ground station for further processing and PDS4 data products were generated and archived under active archive structure. After defined lock in period as per SSPO ISRO HQ, Long Term Archive (LTA) was prepared and interfaced with Chandrayaan-2 Map based browse application hosted from ISSDC for the users. LTA is self-structured, self-documented and peer reviewed data sets. For user's assistance contact information about the team is added under **Annexure III**.

Annexure – I

File Naming Conventions & Formats

This section describes the naming convention used for science data files for the Chandrayaan-2 mission.

ch2_<inst>_<mtc>_<YYYYMMDDTHHMMSSsss>_<p>_<prd>_<stn>.fff

Table 6 shows the details about each parameter used in construction of file name

Table 8 File Naming Conventions

Code	Description
ch2	It denotes the Mission name whose value is fixed as ch2
inst	It denotes the Instrument id, it can have following value = ohr (refers to the OHRC)
M	It denotes the mission phase name, it can have following value = n (refers to the Normal Operations Phase - nop)
T	It denotes the data type; it can have following values: = r (it denotes raw data) = c (It denotes calibrated data)
c	It denotes the imaging mode/camera id/band. It can have following value = p (it denotes panchromatic high resolution camera)
YYYYMMDDTHHMMSSsss	It denotes the observation/imaging start time in format [YYYYMMDDTHHMMSSsss] YYYY = stand for year MM = stand for month in MM format DD = stand for day in DD format T = stand for time. HH = stand for hour in HH format MM = stand for minute in MM format SSS = stand for seconds in SS format ssss = stand for 1/10 millisecond or 100 microsecond in ssss format
P	It denotes the PDS Data Products categories. It value can be any of the following: = d (It denotes data products under data directory) = b (it denotes browse data products under browse directory) = g (it denotes gridded data products under geometry directory)

Prd	It denotes the PDS Data Product Name. It value can be any of the following: = img (It denotes the image data product) = brw (It denotes the browse data product) = grd (It denotes the gridded data product)
Stn	It denotes the station id from which data was acquired. It value can be any of the following: = d32 (ISSDC Bangalore) = d18 (ISSDC Bangalore) = gds (Gold Stone USA) = cnb (Canberra, Australia)
fff	It denotes the standard file type extension. It can have any one of the following values: = img (PDS Image Data File) = xml (PDS Detached Label File) = jpg (PDS Browse Data File) = csv (PDS Geometry Data File)

All underscore characters ‘_’ are mandatory and separate the different fields.

Annexure – II

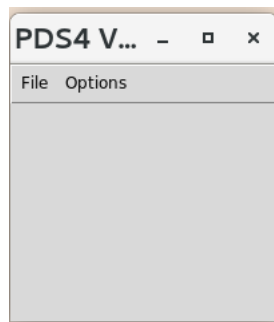
Visualize PDS4 Data Products using PDS4 Viewer

There are many PDS tools are available for opening the Raw/Calibrated Data products.

PDS4Viewer - PDS4Viewer is a simple display tool for visualizing data contained in the PDS4 data standard. PDS4Viewer tool can be downloaded from internet using link “http://sbndev.astro.umd.edu/wiki/PDS4_Viewer”. This Software is available for both Linux as well as window platform. The given data sets can be opened using PDS4Viewer.

The instructions to open the data using PDS4Viewer are as follows:

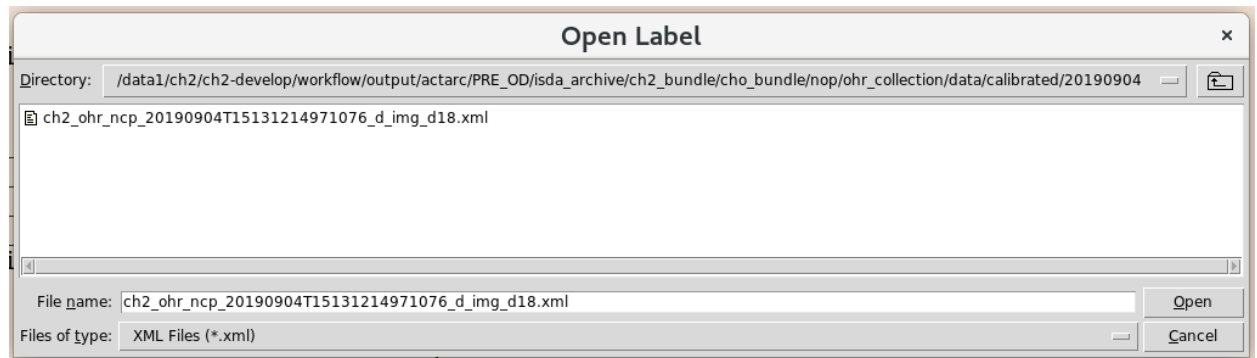
1. Download the PDS4Viewer tool from internet http://sbndev.astro.umd.edu/wiki/PDS4_Viewer. There will be a zip file (for both Linux/Windows) contains the PDS4Viewer.
2. Extract the Zip file into any location.
3. Go to extract the directory, run pds4_viewer. The PDS4Viewer GUI will come which is shown below.



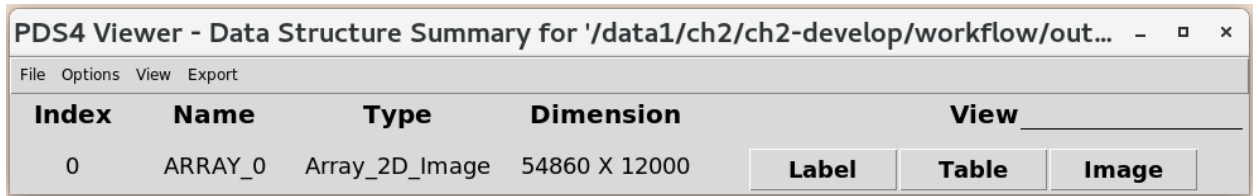
Opening PDS4 Raw/Calibrated Image Product

=====

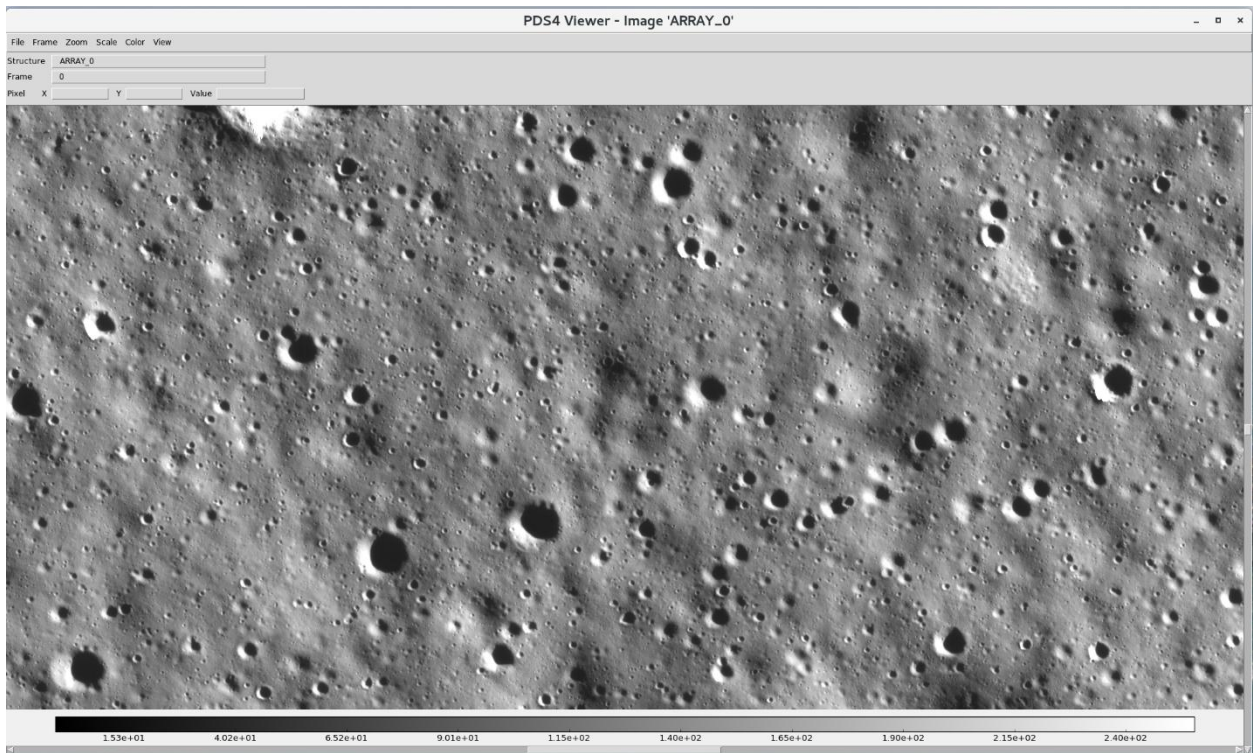
1. After opening the PDS4Viewer display tool. Click on the File menu and select Open Object
2. Click file open dialog will open which is shown below select Label File and click Open button



3. After clicking Open button, the data will be loaded in the PDS4Viewer shown below:



4. For Opening an image, click on the Image button shown on the above figure at step no.
6. Below shows the image from the OHRC Payloads during Decision Support System Operations at MOX on 04 September 2019.



5. For Opening a label, click on the Label by selecting from View drop down menu. Below figures shows the label file as label view and xml view

PDS4 Viewer - Label View

File Edit View

Label

Product_Observational

Identification_Area
logical_identifier: urn:isro:isda:ch2_cho:ohr_calibrated:ch2_ohr_ncp_20190904t15131214971076_d_img_d18
version_id: 1.0
title: Chandrayaan-2 Orbiter OHRC Experiment
information_model_version: 1.11.0.0
product_class: Product_Observational

Modification_History

Modification_Detail
modification_date: 2017-12-22
version_id: 1.0
description: PDS4 product label created by SAC Optical DP Team

Observation_Area

Time_Coordinates
start_date_time: 2019-09-04T15:13:12.14971076Z
stop_date_time: 2019-09-04T15:13:23.278823128Z

Primary_Result_Summary
purpose: Science
processing_level: Calibrated

Investigation_Area
name: Chandrayaan-2
type: Mission

Internal_Reference
lilvid_reference: urn:isro:isda:context:investigation:mission.chandrayaan2::1.0
reference_type: data_to_investigation

Observing_System

Observing_System_Component
name: Chandrayaan 2 Orbiter
type: Spacecraft
description:
Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

Observing_System_Component
name: orbiter high resolution camera
type: Instrument
description:
OHRC (Orbiter High Resolution Camera) is one of the scientific instrument hosted on the Chandrayaan-2 Orbiter. The instrument is a passive electro-optical imaging camera operating in visible-panchromatic band (500-800nm). OHRC images will be used to characterize the landing site (as its primary goal) with a ground sampling of 0.25m

Target_Identification
name: Moon
type: Satellite
description: Moon is a natural satellite of Earth

Mission_Area

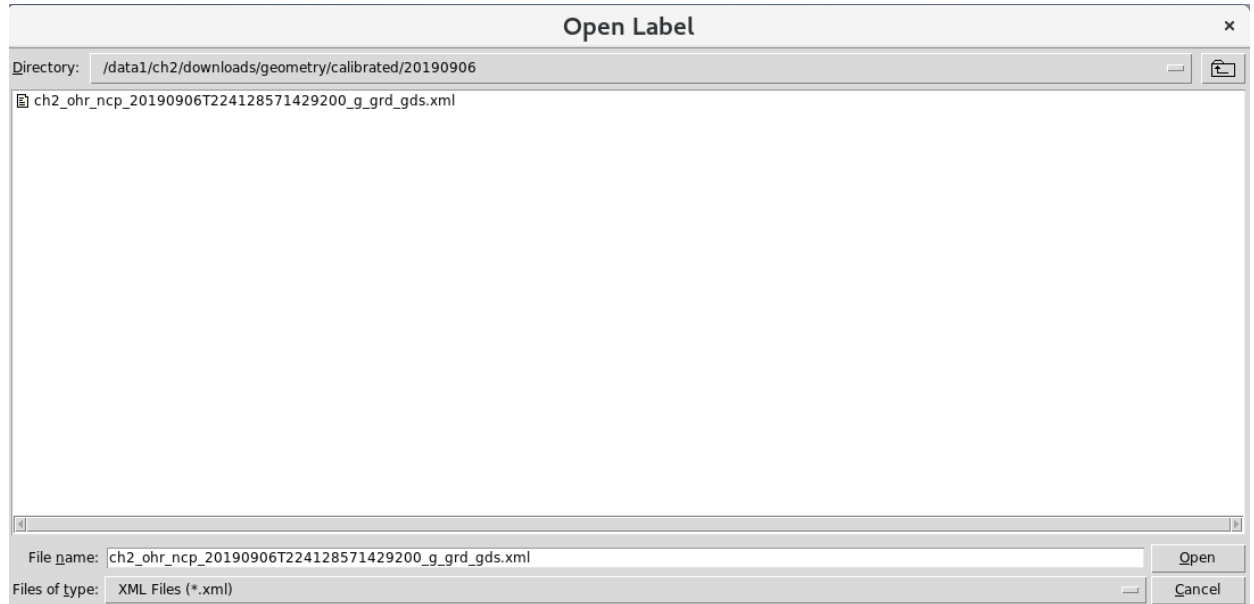
Product_Parameters
imaging_orbit_number: 128
dumping_orbit_number: 129
line_exposure_duration: 178.13999999999999
bits_selection: mid
tdi_stages: 128
detector_pixel_width: 5.2
focal_length: 2080
job_id: OHRXXD18CH00012902NNNN19247164329430_V1_0
level0_dir_name: OHRXXD18CH00012902NNNN19247164329430_V1_0
spacecraft_altitude: 100.54544858552865
reference_data_used: System
orbit_limb_direction: Ascending
spacecraft_yaw_direction: false

Geometry_Parameters

System_Level_Coordinates
upper_left_latitude: -19.516164
upper_left_longitude: -77.143064

Opening PDS4 Geometry Grid Data Product

1. After opening the PDS4Viewer display tool. Click on the File menu and select Open Object
2. Click file open dialog will open which is shown below select Label File and click Open button



3. After clicking Open button, the data will be loaded in the PDS4Viewer shown below:

PDS4 Viewer - Data Structure Summary for '/data1/ch2/downloads/geometry/calibrated/20...						
File Options View Export						
Index	Name	Type	Dimension	View		
1	STREAM_CSV_ID	Table_Delimited	4 cols X 8107 rows	Label	Table	Plot

4. For Opening grid file, click on the table button shown on the above figure at step no. 6. Below shows the grid file from the OHRC Payloads

PDS4 Viewer - Table 'STREAM_CSV_ID'

File Options View

Row #	Longitude	Latitude	Pixel	Scan
0	272.257229	-14.4873495	0	0
1	272.257196	-14.4881259	100	0
2	272.257162	-14.4889024	200	0
3	272.257129	-14.4896788	300	0
4	272.257095	-14.4904554	400	0
5	272.257062	-14.491232	500	0
6	272.257028	-14.4920086	600	0
7	272.256995	-14.4927853	700	0
8	272.256961	-14.493562	800	0
9	272.256928	-14.4943388	900	0
10	272.256894	-14.4951156	1000	0
11	272.256861	-14.4958925	1100	0
12	272.256827	-14.4966694	1200	0
13	272.256794	-14.4974464	1300	0
14	272.25676	-14.4982234	1400	0
15	272.256727	-14.4990005	1500	0
16	272.256693	-14.4997776	1600	0
17	272.25666	-14.5005548	1700	0
18	272.256626	-14.5013321	1800	0
19	272.256593	-14.5021093	1900	0
20	272.256559	-14.5028867	2000	0
21	272.256526	-14.503664	2100	0
22	272.256492	-14.5044415	2200	0
23	272.256459	-14.5052189	2300	0
24	272.256425	-14.5059965	2400	0
25	272.256392	-14.506774	2500	0
26	272.256358	-14.5075517	2600	0
27	272.256325	-14.5083293	2700	0
28	272.256291	-14.5091071	2800	0
29	272.256258	-14.5098848	2900	0
30	272.256224	-14.5106626	3000	0
31	272.256191	-14.5114405	3100	0
32	272.256157	-14.5122184	3200	0
33	272.256124	-14.5129964	3300	0
34	272.25609	-14.5137744	3400	0
35	272.256057	-14.5145525	3500	0
36	272.256023	-14.5153306	3600	0
37	272.255989	-14.5161088	3700	0
38	272.255956	-14.516887	3800	0
39	272.255922	-14.5176653	3900	0
40	272.255889	-14.5184436	4000	0
41	272.255855	-14.519222	4100	0
42	272.255822	-14.5200004	4200	0
43	272.255788	-14.5207789	4300	0
44	272.255755	-14.5215574	4400	0
45	272.255721	-14.522336	4500	0
46	272.255687	-14.5231146	4600	0
47	272.255654	-14.5238933	4700	0
48	272.25562	-14.524672	4800	0
49	272.255587	-14.5254508	4900	0

- For Opening a label, click on the Label by selecting from View drop down menu. Below figures shows the label file as label view and xml view

PDS4 Viewer - Label View
File Edit View

Label

Product_Observational

Identification_Area
logical_identifier: urn:isro:isda:ch2_cho:ohr_calibrated:ch2_ohr_ncp_20190906t224128571429200_g_grd_gds
version_id: 1.0
title: Chandrayaan-2 Orbiter TMC 2 Experiment
information_model_version: 1.11.0.0
product_class: Product_Observational

Modification_History

Modification_Detail
modification_date: 2018-01-04
version_id: 1.0
description: PDS4 product label created by SAC Optical DP Team

Observation_Area

Time_Coordinates
start_date_time: 2019-09-06T22:41:28.571429200Z
stop_date_time: 2019-09-06T22:41:29.635666016Z

Investigation_Area
name: Chandrayaan-2
type: Mission

Internal_Reference
lidvid_reference: urn:isro:isda:context:investigation:mission.chandrayaan2::1.0
reference_type: data_to_investigation

Observing_System

Observing_System_Component
name: Chandrayaan 2 Orbiter
type: Spacecraft
description:
Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

Observing_System_Component
name: orbiter high resolution camera
type: Instrument
description:
OHRC (Orbiter High Resolution Camera) is one of the scientific instrument hosted on the Chandrayaan-2 Orbiter. The instrument is a passive electro-optical imaging camera operating in visible-panchromatic band (500-800nm). OHRC images will be used to characterize the landing site (as its primary goal) with a ground sampling of 0.25m

Target_Identification
name: Moon
type: Satellite
description: Moon is a natural satellite of Earth

File_Area_Observational

File
file_name: ch2_ohr_ncp_20190906T224128571429200_g_grd_gds.csv
local_identifier: STREAM_DELIMITED_CSV_FILE_ID
creation_date_time: 2019-10-29T16:30:23.023
file_size: 273034
records: 8107
md5_checksum: b1caee57f5fbe58d5651c36d9342cd06
comment:
This File contains selenographic coordinates for the image points.
md5_checksum is provided for ensuring data integrity when users are downloading the data.

Header
name: Column headings for Grid Data
local_identifier: HEADER
offset: 0
object_length: 31
parsing_standard_id: PDS DSV 1

Table_Delimited
local_identifier: STREAM_CSV_ID

Annexure – III

Team: Contact Information

Teams	Person Name	Contact	Email Id
Operations Team @ ISSDC IDSN Bylalu, Bengaluru	Sri. Sreenath	080-2809-4416/17/18 (SCC), 2809-4419 (MOX), 2202-9173/74 (ISSDC-IDSN)	sreenath@istrac.gov.in
Principal Investigator (PI) Team @ SAC Ahmedabad	Sri. Aditya Kumar Dagar	07926914309	adagar@sac.isro.gov.in dagar_aditya@yahoo.com